

SOFTWARE CONSTRUCTION AND DEVELOPMENT

Engr. Muhammad Umer Haroon

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Introduction

Course/Class Expectations

- Phones silent and away · No gum chewing
- Arrive on time · Respect everyone
- Minimum 75% attendance
- Submit assessments on time
- No cheating or plagiarism

- What is Software ?
- What is Programming ?
- What is code ?

- Software Engineering emphasizes not to code .
- Sounds contradictory ?
- But to engineer software .

- Is it a theoretical subject ?

Is construction and development same ?

- Software construction refers to the detailed creation of software through coding, debugging, and testing. It involves transforming software design into executable code and integrating individual software components into a functioning whole.
- Software construction is a core activity in most software development life cycles, following design and requirements analysis. It focuses on practical considerations and constraints in building working software from blueprints.

- Key activities in software construction include coding, following best practices and standards, managing dependencies, integrating components, verifying code quality, and debugging.
- Construction may involve creating new code, reusing existing code, or both.
- The goal is to produce high-quality, maintainable software that meets requirements and design specifications. Careful construction is crucial for creating secure, reliable software.

SOFTWARE CONSTRUCTION FUNDAMENTALS

- minimizing complexity
- anticipating change
- constructing for verification
- Reuse
- standards in construction.

The first four concepts apply to design as well as to construction.

- Software construction - > code - > clean code

Michael Feathers, author of *Working Effectively with Legacy Code*

I could list all of the qualities that I notice in clean code, but there is one overarching quality that leads to all of them. Clean code always looks like it was written by someone who cares. There is nothing obvious that you can do to make it better. All of those things were thought about by the code's author, and if you try to imagine improvements, you're led back to where you are, sitting in appreciation of the code someone left for you—code left by someone who cares deeply about the craft.



Naming

- Naming of what ? Choosing good names takes time but saves more than it takes.
- So take care with your names and change them when you find better ones. Everyone who reads your code (including you) will be happier if you do

- The name of a variable, function, or class, should answer all the big questions. It should tell you why it exists, what it does, and how it is used.
- If a name requires a comment, then the name does not reveal its intent. E.g ?????

- int d; // elapsed time in days

int d;

- The name d reveals nothing. It does not evoke a sense of elapsed time, nor of days.
- We should choose a name that specifies what is being measured and the unit of that measurement.

```
int elapsedTimeInDays;
 int daysSinceCreation;
 int daysSinceModification;
 int fileAgeInDays;
```

LIST , TUPLE , DICTIONARY

- List and tuple operations
- How to add , append, remove ,delete ,sort ,reverse, replace ,typecast ,concatenate

For queries

Office : B05, RSB

Counselling Hours :

Monday	11:00 am -12:00 pm
Tuesday	12:00 pm -1:00 pm
Thursday	10:15 am -11:15 am

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